

CAN VISION-LANGUAGE MODELS LEARN TO EXECUTE CLASSICAL SEARCH ALGORITHMS?

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Paper under double-blind review

ABSTRACT

Process supervision can teach vision-language models executable structural search. Prior work found structural enumeration flat while operational reasoning improved under scaling and prompting. A Search Process Policy emits Typed Search Operations carrying every exploration-determining operand. A Trusted Search Runtime validates and applies emissions under bounded Search Memory, without supplying or repairing operands. We measure advantage over an oracle-assisted, validity-constrained random-valid control sampling only runtime-valid operations. On a frozen 15-task text-state development panel for best-first width search, one fixed-seed process-supervised run reached 0.933 success, versus 0.467 for random-valid, 0.0 for the pretrained base, and 1.0 for exact execution, an absolute gain of 0.467 with a paired whole-instance bootstrap lower bound of 0.2. On breadth-first and two additive best-first panels, training matched but did not exceed random-valid at 100%, suggesting learned advantage depends on control headroom. In a 24-task, 12-domain baseline across text, visual, and multimodal states, process supervision reached 125/288 successes versus 240/288 for random-valid, all in additive best-first cells. Breadth-first and width-based cells failed in every modality. Model-generated successors collapsed goal success from 45/45 to 1/45, and two DAgger iterations showed no advantage. These development-stage, single-seed results leave held-out primary evaluation, cross-backbone outcomes, generalization, and transfer unmeasured. Executable search is trainable, and learned control advantage can appear, but remains algorithm- and headroom-dependent and does not yet establish general planning efficacy.

1 INTRODUCTION

Classical planning requires finding a sequence of actions that transforms an initial state into a goal state under specified constraints—a search problem whose formal structure allows instances to be generated systematically, solutions verified automatically, and difficulty controlled precisely, making it a natural probe of reasoning in foundation models (Valmeekam et al., 2023; Stein et al., 2025; Kokel et al., 2026). The formal model behind each instance fixes the task semantics, so errors reflect reasoning rather than ambiguity in the problem statement. The explicit state space also supports two levels of question: how actions apply locally, and how states connect globally. These properties let an evaluator vary difficulty along controlled dimensions and check each answer mechanically.

A recent study used this setting to measure latent planning competence in off-the-shelf language models (Huang et al., 2026). It separated two competencies that did not compensate for each other. Operational reasoning covers local applicability, progression, and validation: whether an action is legal in a state, what state it produces, and whether a goal condition holds. Structural enumeration covers goal reachability, landmarks, and other global state-space structure. Across model scales, and under chain-of-thought prompting, targeted prompting, and scratchpad or agent harnesses, operational reasoning improved, but these gains were not accompanied by comparable structural gains. The evidence was item-level and text-only, drawn from models used as released, without task-specific training.

Structural enumeration is related to global search-space reasoning, but it is not identical to full search execution. Identifying which states are reachable, or which serve as landmarks, does not require controlling a frontier across a budgeted episode. The prior study established this profile

observationally. It did not test whether the structural competency can be acquired through training, or whether a model can sustain a declared procedure over a complete episode. Closing that gap calls for direct intervention: train the model to make the exploration decisions of a classical search algorithm, then measure whether structural/process competence changes.

Our intervention replaces direct plan generation with learned execution of a declared search algorithm. A Search Process Policy is trained to emit Typed Search Operations, discrete steps such as expanding a state, generating a successor, updating the frontier, or applying a goal test. A Trusted Search Runtime validates and applies each emission, checks Algorithm Invariants on every operation, and maintains bounded external Search Memory holding frontier, visited, novelty, and priority state. Ownership is replay-determinative. The emission must contain every operand that determines exploration, and the runtime may not select omitted operands, reorder candidates for the policy, or silently repair an invalid emission. Replaying the raw emissions against the same task and Search Memory must reproduce the explored trace. Operation validity, invariant compliance, and episode success are measured separately from learned control advantage, which is computed against an oracle-assisted, validity-constrained random-valid control that samples only runtime-valid operations.

The study asks whether a policy can learn to execute declared algorithms with valid, invariant-compliant traces, whether learned execution yields control advantage over the random-valid control, and whether the acquired capability transfers to planning benchmarks and broader search-intensive reasoning. The active algorithms are BFS, a complete unpruned width-based best-first search (BFWS), and two additive best-first variants, weighted and greedy. Process supervision on search-trace segments is the core training condition, and two-iteration DAgger, model-generated successors, and curriculum-by-modality have also been executed. Modality conditions present the same task content as text-state, visual-state, or multimodal-state observations under equal Search Memory capacity. References are the pretrained base, the random-valid control, and exact classical execution. Generalization and robustness checks, second-backbone evaluation, transfer, and the authorized final primary evaluation remain unmeasured.

The results form a headroom-dependent map. On a frozen 15-task text-state development panel under BFWS, one fixed-seed process-SFT run achieved 0.933 success, versus 0.467 for random-valid, 0.0 for the base, and 1.0 for exact execution. This is the only clear positive learned-control comparison, and it is development-stage. On BFS and the two additive best-first development panels, process SFT matched random-valid at 100%, so saturation left no room for measured advantage. An expanded baseline of 24 tasks across 12 domains and all three modalities reinforces the boundary: process SFT succeeded on 125 of 288 instances versus 240 for random-valid, 0 for the base, and 288 for exact execution, with every success in additive best-first cells, while BFS and width-based best-first failed on all 24 tasks in every modality. Two DAgger iterations added no advantage, and replacing trusted successors with model-generated ones dropped goal success from 45/45 to 1/45. All of these findings are single-seed and development-stage. The held-out primary evaluation, generalization, second-backbone outcomes, and transfer remain unmeasured.

This paper makes three contributions.

- **From diagnosis to intervention.** We execute the training-based follow-up to a diagnosed structural bottleneck: rather than prompt around flat structural enumeration, we train models to execute declared search algorithms and map where learned control emerges.
- **A replay-determinative policy/runtime interface.** Typed Search Operations carry every exploration-determining operand, while the no-repair Trusted Search Runtime validates and applies them. Operation validity, invariant compliance, episode success, and learned control advantage remain separate measurements.
- **A headroom-dependent empirical map.** Across development panels, learned control advantage appears only where saturation leaves headroom: a positive BFWS comparison, ceiling-matched BFS and additive best-first results, cross-algorithm and cross-modality failures in an expanded baseline, and negative DAgger and model-generated-successor boundaries. This is development-stage evidence, not final efficacy.

The paper therefore maps conditional, development-stage learnability of executable search rather than establishing general planning efficacy, with the final held-out evaluation, second-backbone outcomes, generalization, and transfer still unresolved.

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A APPENDIX

The Appendix will retain the complete evidence contract, including operation schemas, Algorithm Invariant definitions, runtime pseudocode, data and split procedures, receipt tables, statistics, adapters, and narrowly scoped feasibility details of retained infrastructure.